

Chapter 1

Partition of the Plane

1.1 Plane Separation Axiom

The previous axioms of betweenness govern the linear ordering of points and are therefore limited to a single dimension. To extend this to two dimensions, we introduce the Plane Separation Axiom. This asserts that a line acts as a partition splitting the plane into two disjoint, non-overlapping regions, which is then used to describe a notion of points membership to one side of a given line through its logical equivalence to the negation of the previously defined notion of segment intersection.

Axiom 1 (Plane Separation). *For every line $l \in \mathbf{L}$, there exist two sets $H_{l,A}, H_{l,B} \subset \mathbf{P} \setminus l$ such that:*

1. $H_{l,A} \cup H_{l,B} = \mathbf{P} \setminus l \wedge H_{l,A} \cap H_{l,B} = \emptyset$.
2. $\forall A, B \in \mathbf{P} \setminus l: (\exists X \in l, \mathcal{B}(A, X, B) \iff \{A, B\} \not\subset H_{l,1} \wedge \{A, B\} \not\subset H_{l,2})$.

explain the meaning-of-axioms-here

1.2 Same-Side Relation

We formalize the concept of 'being on the same side' through the Same-Side Relation ($\mathcal{S}$). It is defined by the absence of a separation/partition of the plane: two points are related if and only if unique segment connecting them does not contain any point of the boundary line l . This definition ensures the relation is based upon the primitive notion of betweenness. Specifically, we define $\mathcal{S}$ as a subset of the cartesian product $\mathcal{S} \subseteq \mathbf{P} \times \mathbf{P} \times \mathbf{L}$.

Definition 1 (Same-Side Relation). *Given a line $l \in \mathbf{L}$, any two points $A, B \in \mathbf{P} \setminus l$ are related by $\mathcal{S}$ if there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, B)$. That is,*

$$\mathcal{S}(A, B, l) \iff \nexists X \in l: \mathcal{B}(A, X, B).$$

This geometric object allows us to move beyond comparisons between single points. A point X being on the same side as point Y is the same as X being an element of the half-plane $H_{l,y}$. This will show incredibly useful in defining derived geometric figures.

1.3 Theorems

Theorem 1 (Same-Side Reflexivity). *For any point $A \in \mathbf{P}$ that is not a member of any given line $l \in \mathbf{L}$, the following must hold:*

$$\forall l \in \mathbf{L}, \forall A \in \mathbf{P}: \mathcal{S}(A, A, l).$$

Proof. Let there be a line $l \in \mathbf{L}$ and a point $A \in \mathbf{P} \setminus l$. We proceed by contradiction. Assuming the same-side relation $\mathcal{S}(A, A, l)$ is false, it follows that $\nexists X \in l: \mathcal{B}(A, X, A)$ is false by definition, and that therefore there does exist an $X \in l$ satisfying these conditions. As a consequence, by the Identity/Reflexivity of Betweenness $A = X$.

However, we have defined that $A \notin l$. Since our assumption leads to a contradiction, it must be that no such point X exists. Thus, $\mathcal{S}(A, A, l)$ is true. $\square$

Theorem 2 (Same-Side Symmetry).

$$\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l).$$

Proof. Firstly, we note that $\mathcal{S}(A, B, l) \iff \nexists X \in l: \mathcal{B}(A, X, B)$ by definition. By the Symmetry of Betweenness, the right-hand side of this is equivalent to $\nexists X \in l: \mathcal{B}(B, X, A)$, thus showing

$$X \in l: \neg \mathcal{B}(A, X, B) \iff X \in l: \neg \mathcal{B}(B, X, A).$$

The hand-side is by definition equivalent to $\mathcal{S}(B, A, l)$. Thus, by the transitivity of the biconditional, we conclude $\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l)$. $\square$

Theorem 3 (Same-Side Transitivity). *Given any line $l \in \mathbf{L}$ and points $A, B, C \in \mathbf{P} \setminus l$, the relation $\mathcal{S}$ satisfies:*

$$\mathcal{S}(A, B, l) \wedge \mathcal{S}(B, C, l) \implies \mathcal{S}(A, C, l).$$

Proof. By the Plane Separation Axiom, we know there must exist two distinct sets H_{l1} and H_{l2} whose union is $\mathbf{P} \setminus l$. Since A, B , and C are not on l , each must belong to exactly one of these two sets.

Assume $\mathcal{S}(A, B, l)$. By definition, they are contained within the same set. Without loss of generality, let $A, B \in H_{l1}$.

Assume $\mathcal{S}(B, C, l)$. Similarly, this implies they are in the same set. We previously established that $B \in H_{l1}$, implying that $C \in H_{l1}$ by the plane separation axiom ($H_{l1} \cap H_{l2} = \emptyset$).

We now have $A \in H_{l1}$ and $C \in H_{l1}$. By the inverse of PSA2, there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, C)$. Thus, by definition, A and C must be related by the same-side relation. That is, $\mathcal{S}(A, C, l)$. $\square$

With the reflexivity, symmetry, and transitivity of the same-side relation demonstrated, it follows by definition that it is an equivalence relation. This result may seem trivial, but it allows for application of known properties of equivalence relations. One such property is *equivalence classes*, which are the subsets for which the relation holds. For our geometry, this partitions the plane into two *half-planes*.

Theorem 4 (Partition Theorem). *Any pair of half-planes alongwith the line forming them together cover the plane.*

Proof. First note that the relation form $\square$

1.4 Half-Planes

By grouping all points related to a reference point A by $\mathcal{S}$, we construct the geometrical object known as the half-plane $(H_{l,A})$. This transition allows us to treat regions of the plane as sets that can be intersected to form more complex figures, such as angles and triangles.

Definition 2 (Half-Plane). *Let $l \in \mathbf{L}$ be a line and let $A \in \mathbf{P}$ be a reference point such that $A \notin l$. The half-plane determined by l and A , denoted $H_{l,A}$, is the subset of the point-set consisting of all points on the same side of l as A . That is,*

$$H_{l,A} := \{X \in \mathbf{P} \setminus l \mid \mathcal{S}(A, X, l)\}.$$

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